Research questions and hypothesis

INFO B302 – Introduction à la démarche scientifique (IDS)

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What is: the State-of-the-art

…things that were… things that are…

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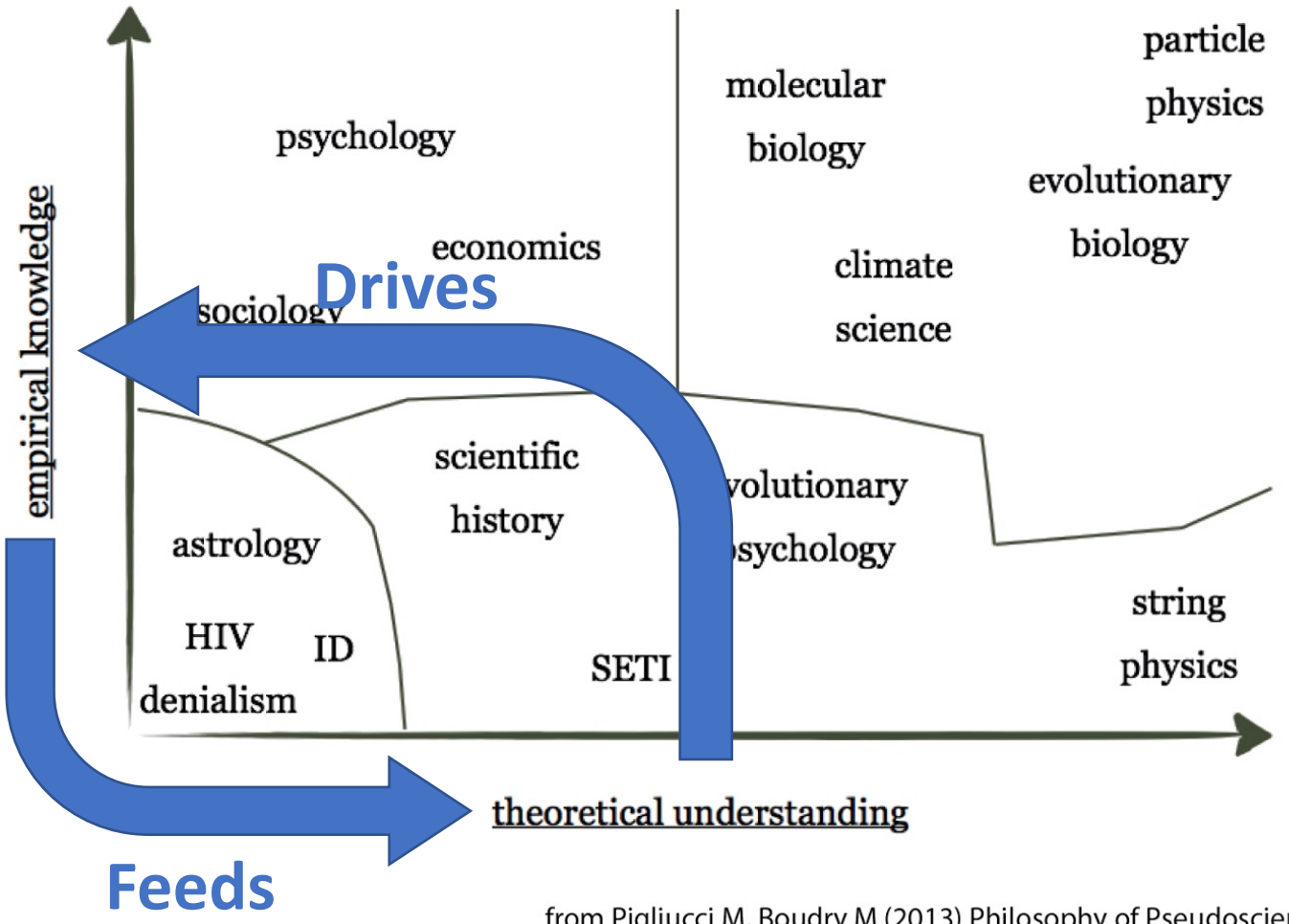
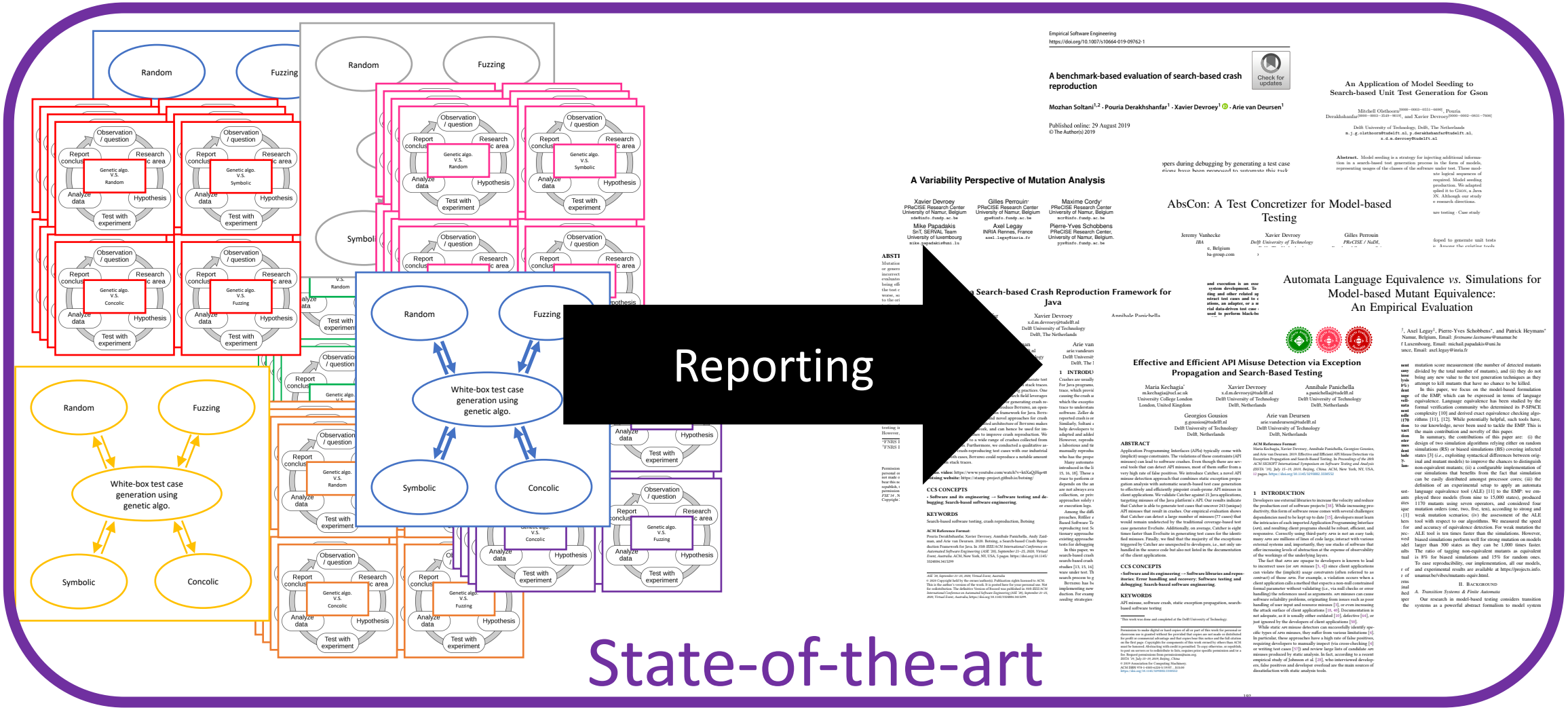
The demarcation problem



from Pigliucci M, Boudry M (2013) Philosophy of Pseudoscience: Reconsidering the Demarcation Problem; University of Chicago Press. with permission of M Pigliucci.

Reporting in (peer-reviewed) publications



Systematic Literature Review

1. Plan the review

1. Specify research questions
2. Develop review protocol
3. Validate review protocol

2. Conduct the review

1. Identify relevant research
2. Select primary studie
3. Assess study quality
4. Extract required data
5. Synthesise data

3. Document the review

1. Write review report
2. Validate report

What could be: Research questions

and some things…that have not yet come to pass.

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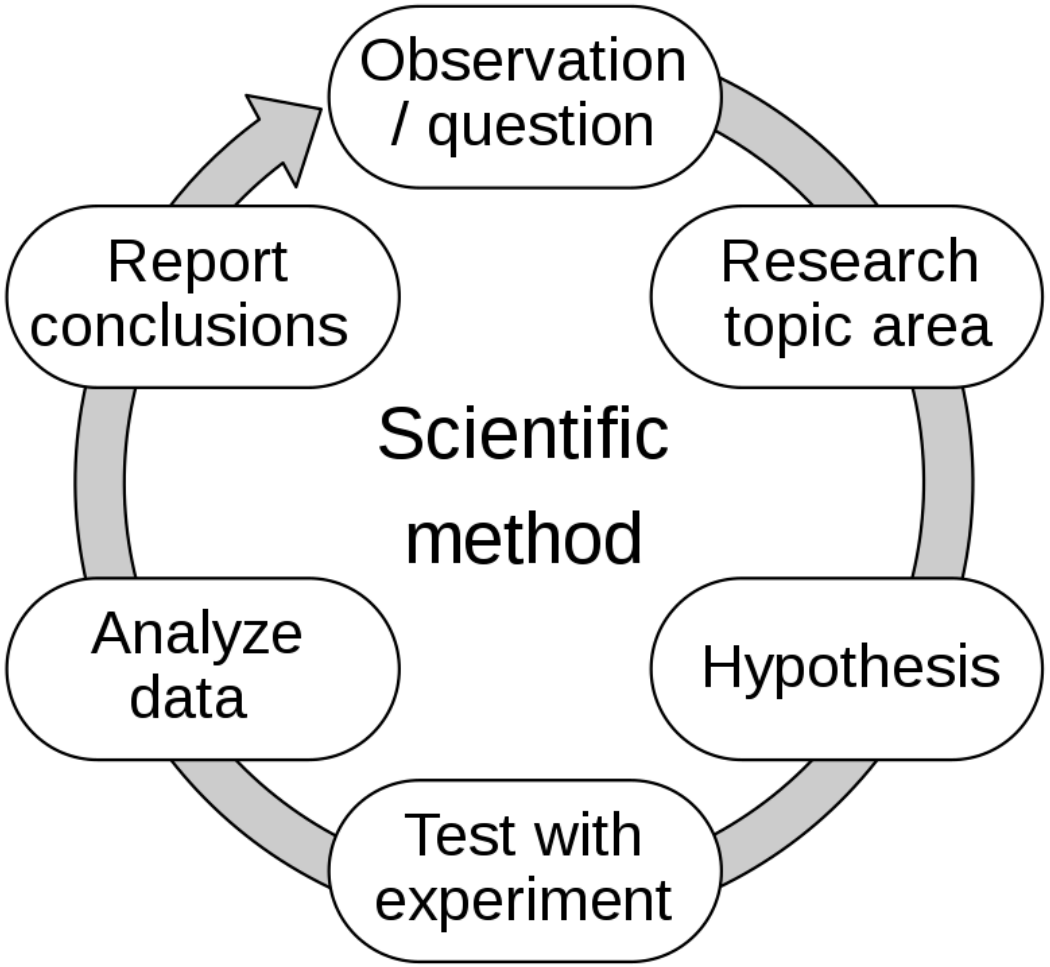
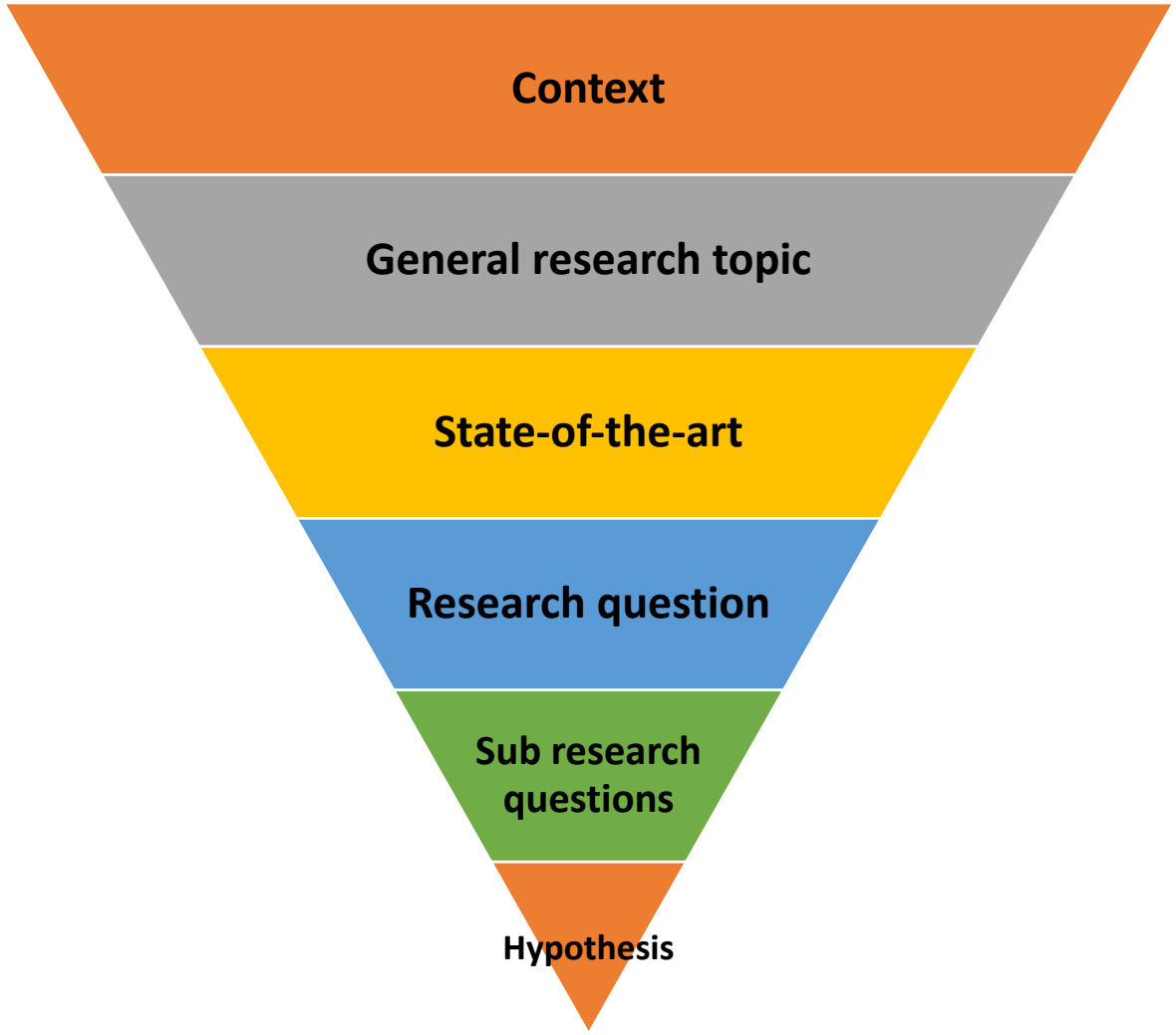
How science works (roughly)

- It all starts with a research question
- Focuses on the topic under research
- Based on the state-of-the-art
- Sets the scope the study
- Drives the research protocol
- Is answered in the conclusion



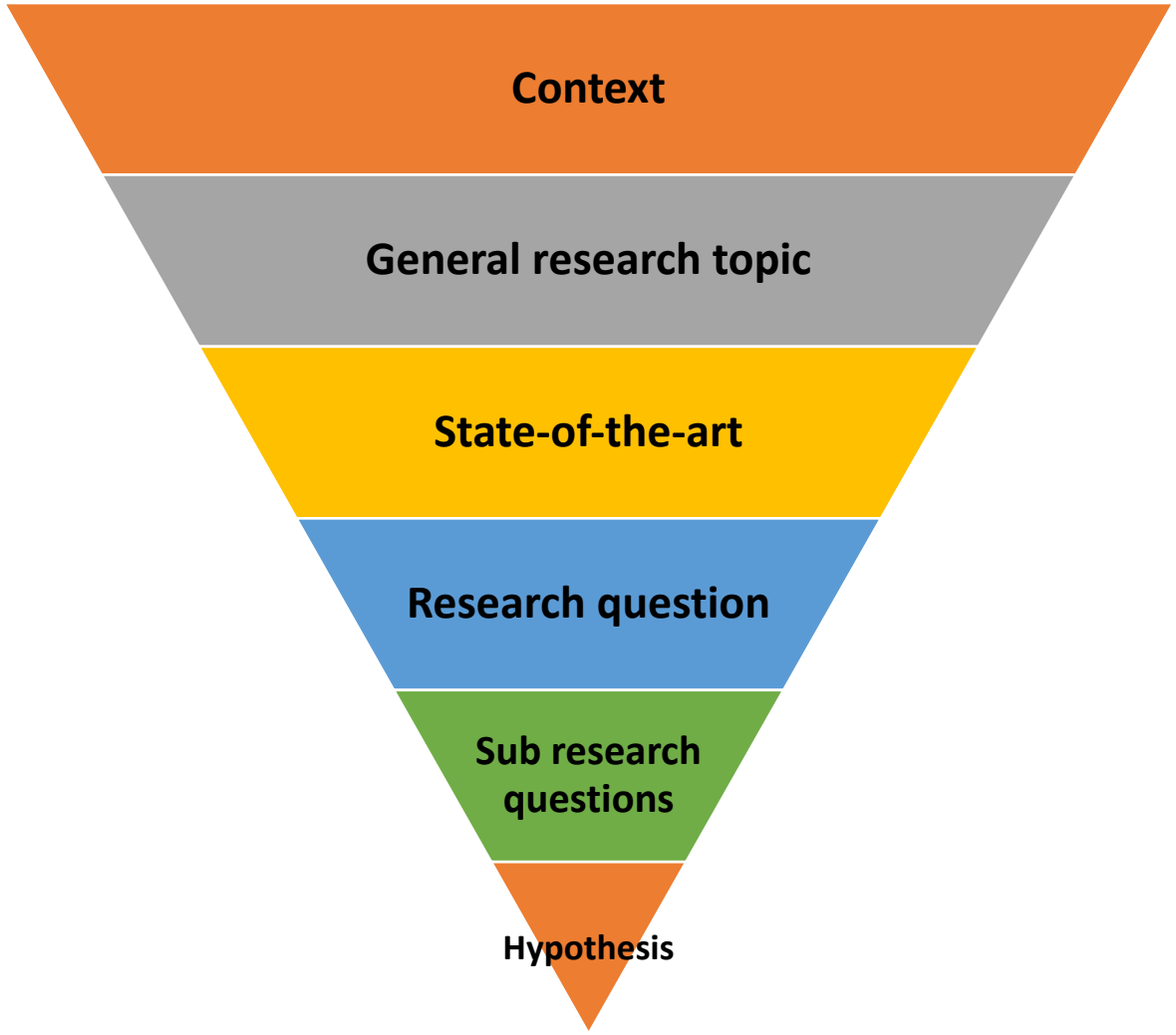
Building a research question

- Context and general topic
  - Automated test case generation
- Check the state-of-the-art
  - Search-based software testing, Fuzzing, Symbolic testing, etc.
- Narrow down to a specific niche to focus on
  - Search-based sofware testing for Java programs



Building a research question

- Identify a research problem to address
  - Evaluation of a new search-based test generation approach XYZ for Java
- Formulate the research question
  - How does the approach XYZ compares to the state-of-the-art regarding its fault-finding capabilities?



Examples of research questions

- Describing and exploring
  - What are the characteristics of X (in context Y)?
  - How has X evolved over time (in context Y)?
  - What are the main factors in X?
  - How has X dealt with Y?
- Explaining and testing
  - What is the relationship between X and Y?
  - What is the role of X in Y?
  - What is the impact of X on Y?
  - How does X influence Y?
  - What are the causes of X?

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Examples of research questions

- Evaluating and acting
  - What are the advantages and disadvantages of X (in context Y)?
  - How effective is X (for task Y)?
  - How can X be achieved (in context Y)?
  - What are the most effective strategies to improve X (in context Y)?
  - How can X be used in Y?

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Focused and researchable

- Focuses on a single topic and problem
  - Defines the scope of the research
  - Example
    - 🤬 What testing approaches exist?
    - 😐 What testing approaches exist for crash reproduction?
    - 😍 What automated testing approaches can be used for crash reproduction?
- Answerable using primary or secondary data
  - It should be possible to define a protocol to answer the question
  - Relying either on qualitative date, quantitative data, or existing studies

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Focused and researchable

- Clearly indicates criteria used
  - Avoid subjective words like *good*, *bad*, *better* and *worse*
  - Use a measurable definition
  - Example
    - 🤬 Is XYZ better than the state-of-the-art for test case generation?
    - 😐 How does the approach XYZ compares to the state-of-the-art?
    - 😍 How does the approach XYZ compares to the state-of-the-art regarding its fault-finding capabilities?

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Feasible and specific

- Answerable within practical constraints
  - Defines the (practical) scope of the research
  - Time, resources, access to data, etc.
- Uses specific, well-defined concepts
  - Avoid vague language and broad ideas
  - Clearly indicates what, who, where and when it addresses
  - Example
    - 😡 What effect does social media have on people's minds?
    - 😐 What effect does daily use of social media have on attention span?
    - 😍 What effect does daily use of Twitter/Instagram/9Gag/… have on the attention span of bachelor students?

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Complex and arguable

- Cannot be answered with yes or no
  - No space for investigation and discussion
  - Can <anything/anybody do anything> ?
  - (this will most likely get you a 0/20 for your grade)
- Cannot be answered with easily found facts and figures
  - If the answer is on Wikipedia, the question is not worth your time
  - A research question should require original data, synthesis of multiple sources, interpretation and argument to provide an answer
- Provides scope for debate and deliberation
  - Results should be interpreted

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Relevant and original

- Addresses a problem relevant to your field or discipline
  - Focus on a gap in the existing knowledge
    - Including replication of results
  - (hence the importance of a good bibliographic search)
- Contributes to a topical social or academic debate
  - Should aim to contribute to an existing debate
  - Should produce knowledge that future researchers or practitioners can build on
  - What about fundamental research?
- Has not already been answered
  - Should have some novelty
    - Taking a new angle, focusing on a specific aspect, etc.

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Research question construction: PICOT

- Commonly used in clinical research and evidence-based studies
- Patient, population or problem
  - Who or what is the patient, population or problem in question?
- Intervention
  - What is the intervention (action or treatment) being considered?
- Comparison or control
  - What other interventions should be considered?
- Outcome or objective
  - What is the desired or expected outcome or objective?
- Time frame
  - How long will it take to reach the desired outcome?

https://onlinenursing.duq.edu/blog/formulating-a-picot-question/

PICOT: Example

- Patient, population or problem
  - Java classes
- Intervention
  - Generate unit tests using XYZ
- Comparison or control
  - Generating unit tests using A, B, and C
- Outcome or objective
  - Compare fault-finding capabilities
- Time frame
  - Time budgets of 1, 3, and 5 minutes
- How does the approach XYZ compares **to A, B, and C** regarding its fault-finding capabilities **with time budgets of 1, 3, and 5 minutes**?

https://onlinenursing.duq.edu/blog/formulating-a-picot-question/

Checklist: FINER

- Feasible
  - Is it in the researcher's ability to investigate the research question?
- Interesting
  - Is the research question interesting to the researcher, the peers, and the community?
- Novel
  - Does answering the question bring new insights or confirm previous results?
- Ethical
  - Does answering the research question respect the researcher's and their institution ethical and legal standards?
- Relevant
  - Is the research question relevant to the scientific community (and the larger public)?

Sub-questions and hypothesis

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Sub-research questions

- Refine a research question to clarify the scope
- Lift any remaining ambiguities
- Example: How does the approach XYZ compares to the state-of-the-art regarding its fault-finding capabilities?
  - Fault-finding capabilities = coverage + mutation score
    - How does the approach XYZ compares to the state-of-the-art regarding the **line coverage** of the generated tests?
    - How does the approach XYZ compares to the state-of-the-art regarding the **mutation score** of the generated tests?

Sub-research questions

- Example: How does the approach XYZ compares to the state-of-the-art regarding its fault-finding capabilities?
  - Fault-finding capabilities = real faults
    - How does the approach XYZ compares to the state-of-the-art regarding the **number of faults detected by** the generated tests?
  - Fault-finding capabilities = ability of the developers to identify faults easily when a test fails
    - How does the approach XYZ compares to the state-of-the-art regarding the **time required by developers to identify the root cause of a failing test**?

Hypothesis

- Refine a research question to clarify the scope
- Alternative (or complement) to sub-research questions
- Example: How does the approach XYZ compares to the state-of-the-art regarding its fault-finding capabilities?
  - XYZ produces test with a higher line coverage than A, B, and C
  - XYZ produces test with a higher mutation score than A, B, and C
  - Developers relying on XYZ require less time than developers relying on A, B, or C to identify the root cause of a test failure

Hypothesis

- Can be used to drive the data analysis
- Statistical analysis of quantitative data to reject a hypothesis
  - Arcuri, Andrea, and Lionel Briand. "A hitchhiker's guide to statistical tests for assessing randomized algorithms in software engineering." Software Testing, Verification and Reliability 24.3 (2014): 219-250. https://doi.org/10.1002/stvr.1486

D3 Contexte et question de recherche

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Structure

- Contexte et objectifs
  - Similaire à ce que l’on trouve dans un résumé structuré
- Une ou deux question(s) de recherche
- Pour chaque question de recherche
  - Sous-questions ou hypothèses
  - Une première description générale de la manière dont la question sera traitée
- Pensez à valider vos questions de recherche

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NEWS

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Le FNRS signe **la déclaration de DORA**

DORA

La déclaration de DORA ou « *San Francisco Declaration on Research Assessment* » (www.sfdora.org) consiste en un ensemble de recommandations relatives aux bonnes pratiques en matière d'évaluation de la recherche scientifique. Pour les agences de financement comme le FNRS, il s'agit notamment d'accorder moins de poids dans l'évaluation aux noms et à l'*Impact Factor* des revues dans lesquelles les travaux des chercheurs sont publiés. En effet, l'impact moyen d'une revue, son niveau de prestige, ou son Impact Factor par exemple, ne présagent pas forcément de la qualité et de l'impact réel d'un article qui y serait publié, or les *peer reviewers* ont souvent tendance à considérer les noms et *Impact Factors* des revues dans lesquelles les travaux d'un chercheur sont publiés comme révélateurs en soi de la qualité du travail du chercheur en question.

Le FNRS vient donc de signer cette déclaration. L'occasion d'apporter une modification dans l'appel Bourses et Mandats 2022 : les candidats Chargés de recherches et Chercheurs qualifiés ont été invités à joindre à leur dossier de candidature les 5 publications les plus représentatives de leur carrière (qui devaient jusqu'ici être seulement listées et justifiées dans la partie scientifique du dossier de candidature).

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